EVERY FPPF COVERING IS REFINED, ETALE-LOCALLY ON THE BASE, BY SURJECTIVE FINITE LOCALLY FREE MAPS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13607-13617 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 13619-13646, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Middle member of the chapter's fppf-domination trio (the other two are this wave's L16108 dominate-fppf-integral predecessor dominate-fppf-finite and the ph card collected last wave); the etale-local form is the one used by the ph topology comparisons.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemmas qf-fp-flat-dominates-fppf (reducing fppf covers to locally quasi-finite ones) and etale-makes-quasi-finite-finite-at-point (the etale-local finiteness-at-a-point construction); Morphisms-chapter stability lemmas for flat finitely presented base change and the finite-flat-implies-finite-locally-free criterion.

Source reference. Original lemma 02LK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11924–11934. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0572: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6466–6474. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01TS: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13646–13646. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01U9: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13646–13646. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02KB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13646–13646. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $\{S_i \to S\}_{i \in I}$ be an fppf covering. Then there exist*

(1) *an étale covering $\{S'_a \to S\}$,*
(2) *surjective finite locally free morphisms $V_a \to S'_a$,*

such that the fppf covering $\{V_a \to S\}$ refines the given covering $\{S_i \to S\}$.

Proof. We may assume that each $S_i \to S$ is locally quasi-finite, see Lemma 0572.

Fix a point $s \in S$. Pick an $i \in I$ and a point $s_i \in S_i$ mapping to s. Choose an elementary étale neighbourhood $(S', s) \to (S, s)$ such that there exists an open

$$S_i \times_S S' \supset V$$

which contains a unique point $v \in V$ mapping to $s \in S'$ and such that $V \to S'$ is finite, see Lemma 02LK. Then $V \to S'$ is finite locally free, because it is finite and because $S_i \times_S S' \to S'$ is flat and locally of finite presentation as a base change of the morphism $S_i \to S$, see Morphisms, Lemmas 01TS, 01U9, and 02KB. Hence $V \to S'$ is open, and after shrinking S' we may assume that $V \to S'$ is surjective finite locally free. Since we can do this for every point of S we conclude that $\{S_i \to S\}$ can be refined by a covering of the form $\{V_a \to S\}_{a \in A}$ where each $V_a \to S$ factors as $V_a \to S'_a \to S$ with $S'_a \to S$ étale and $V_a \to S'_a$ surjective finite locally free. □